

BASIC FILE HANDLING IN C++

In a Visual Studio (VS) project folder, the file must already exist for a read operation to take place. In other words, the file needs to exist for a read operation to be successful. However, this is not the case with write operations. If the file is not present, a new file is created when you try to open it using the following syntax: `writeInFile.open("file1.txt");`, where the file name will be "file1.txt".

For writing to a file, you use **ofstream**, and for reading or extracting data from a file, you use **ifstream**. The `>>` operator is used with **ifstream** for input, and `<<` operator is used with **ofstream** for output.

Write in File

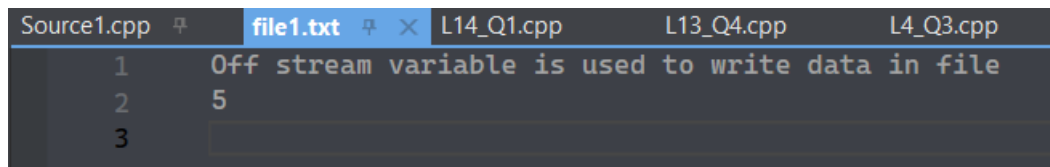
```
#include<iostream>
#include<fstream> //include this to write in file
using namespace std;
int main()
{
    ofstream writeInFile;

    writeInFile.open("file1.txt"); //creating a file if it don't exists

    int var = 5;
    if (writeInFile.is_open())
    {
        writeInFile << "Off stream variable is used to write data in file\n";
        writeInFile << var << endl;
    }
    else
        cout << "file not open\n";

    writeInFile.close();    //closing the file

    return 0;
}
```



Write again in Same file

```
#include<iostream>
#include<fstream> //include this to write in file
using namespace std;
int main()
{
    ofstream writeInFile;

    writeInFile.open("file1.txt"); //creating a file if it don't exists

    int var = 5;
    if (writeInFile.is_open())
    {
        writeInFile << "Off stream variable is used to write data in file\n";
        writeInFile << var << endl;
    }
    else
        cout << "file not open\n";

    writeInFile.close();    //closing the file
}
```

```

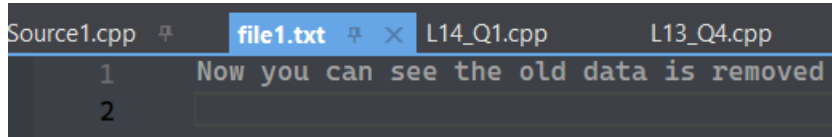
//writing data again in file

writeInFile.open("file1.txt"); //creating a file if it don't exists

if (writeInFile.is_open())
{
    writeInFile << "Now you can see the old data is removed\n";
}
else
    cout << "file not open\n";

writeInFile.close();    //closing the file
return 0;
}

```



Source1.cpp file1.txt L14_Q1.cpp L13_Q4.cpp

```

1 Now you can see the old data is removed
2

```

Using ios::app

```

#include<iostream>
#include<fstream> //include this to write in file
using namespace std;
int main()
{
    ofstream writeInFile;

    writeInFile.open("file1.txt"); //creating a file if it don't exists

    int var = 5;
    if (writeInFile.is_open())
    {
        writeInFile << "Off stream variable is used to write data in file\n";
        writeInFile << var << endl;
    }
    else
        cout << "file not open\n";

    writeInFile.close();    //closing the file

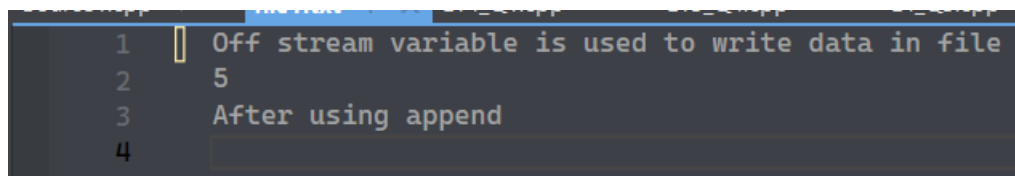
    //writing data again in file

    writeInFile.open("file1.txt", ios::app); //use ios::app to append at the end of
previous file

    if (writeInFile.is_open())
    {
        writeInFile << "After using append\n";
    }
    else
        cout << "file not open\n";

    writeInFile.close();    //closing the file
    return 0;
}

```



```

1 Off stream variable is used to write data in file
2 5
3 After using append
4

```

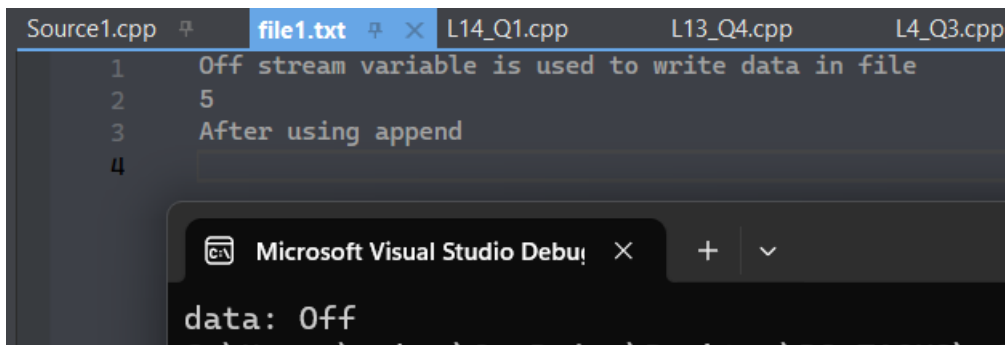
Reading data from file

```
#include<iostream>
#include<fstream> //include this to write in file
using namespace std;
int main()
{
    ifstream readFromFile;

    readFromFile.open("file1.txt"); //opening the file
    string data; //variable to store data
    if (readFromFile.is_open())
    {
        readFromFile >> data; //now the first string is stored in data, jahan tk
        pehli space milli gi wahan tak
    }
    else
        cout << "file not open\n";

    cout << "data: " << data;

    readFromFile.close();
    return 0;
}
```



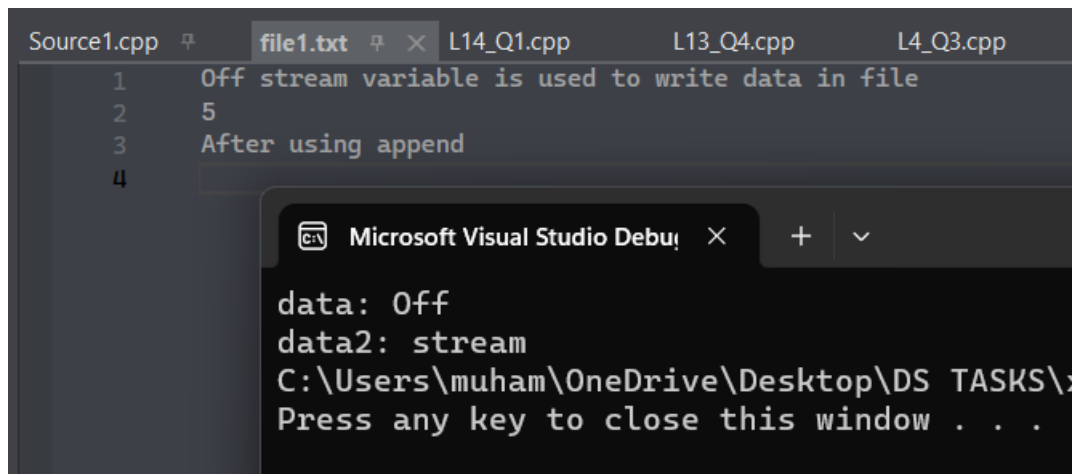
Reading from file

```
#include<iostream>
#include<fstream> //include this to write in file
using namespace std;
int main()
{
    ifstream readFromFile;

    readFromFile.open("file1.txt"); //opening the file
    string data, data2; //variable to store data
    if (readFromFile.is_open())
    {
        readFromFile >> data >> data2; //now the first string is stored in data,
        jahan tk pehli space milli gi wahan tak and 2nd is stroed in data2
    }
    else
        cout << "file not open\n";

    cout << "data: " << data;
    cout << "\ndata2: " << data2;

    readFromFile.close();
    return 0;
}
```



The screenshot shows the Visual Studio IDE with a C++ source file named 'file1.txt'. The code in the file is as follows:

```
1 Off stream variable is used to write data in file
2 5
3 After using append
4
```

The output window, titled 'Microsoft Visual Studio Debug Console', displays the following text:

```
data: Off
data2: stream
C:\Users\muham\OneDrive\Desktop\DS TASKS\
Press any key to close this window . . .
```

Reading whole line from file

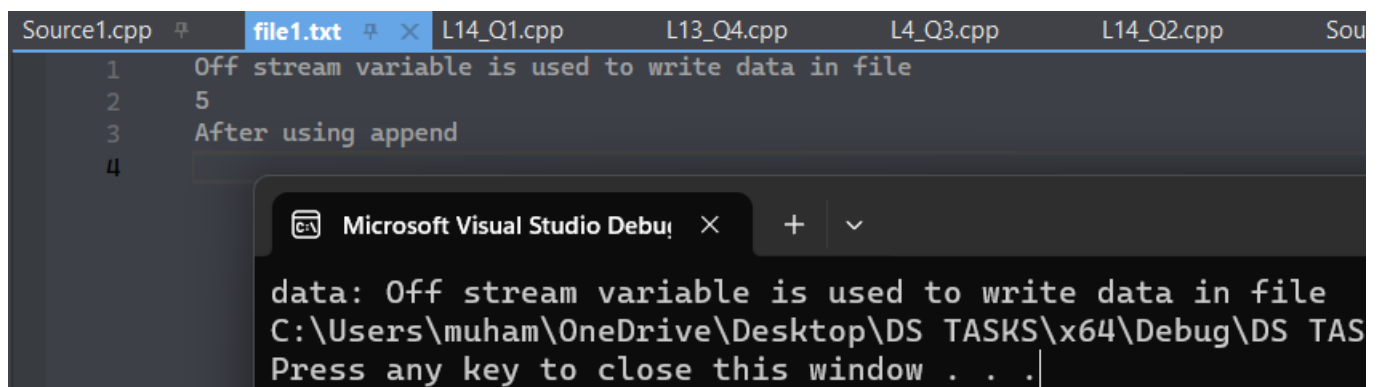
```
#include<iostream>
#include<string> // used to getline
#include<fstream> //include this to write in file
using namespace std;
int main()
{
    ifstream readFromFile;

    readFromFile.open("file1.txt"); //opening the file
    string data; //variable to store data
    if (readFromFile.is_open())
    {
        getline(readFromFile, data);    //used to get whole line from file in the data
        variable
    }

    else
        cout << "file not open\n";

    cout << "data: " << data;

    readFromFile.close();
    return 0;
}
```



This screenshot shows the same Visual Studio IDE setup as the first image, but with an additional tab 'L14_Q2.cpp' and a different output. The source code in 'file1.txt' remains the same. The output window now displays:

```
data: Off stream variable is used to write data in file
C:\Users\muham\OneDrive\Desktop\DS TASKS\x64\Debug\DS TAS
Press any key to close this window . . .|
```

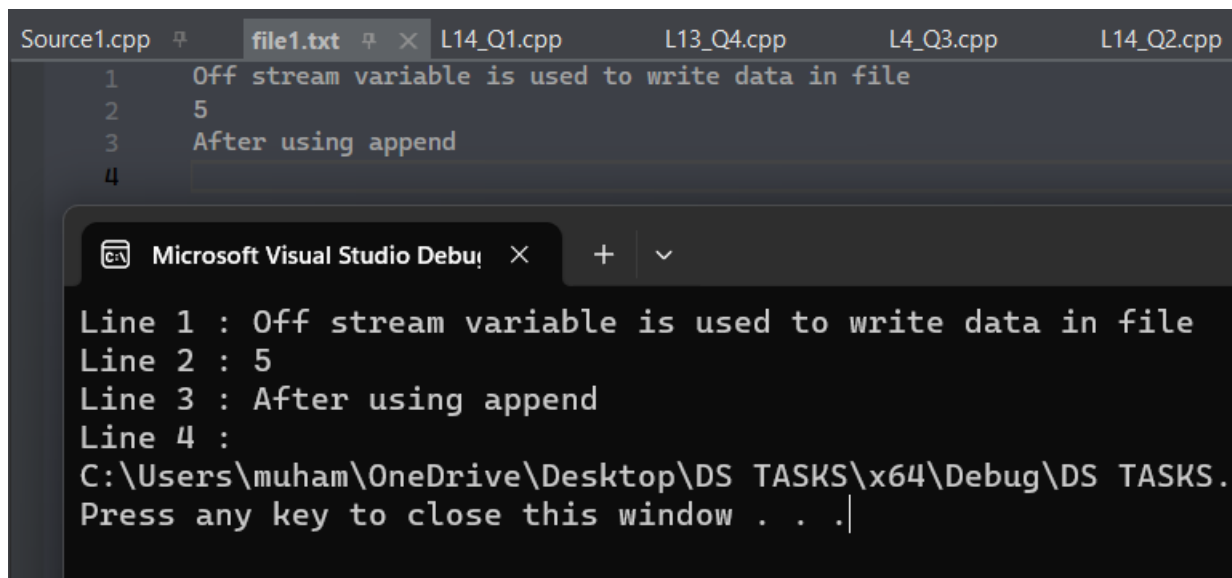
Reading whole file method 1

```
#include<iostream>
#include<string> // used to getline
#include<fstream> //include this to write in file
using namespace std;
int main()
{
    ifstream readFromFile;

    readFromFile.open("file1.txt"); //opening the file
    string data; //variable to store data
    int i = 0;
    if (readFromFile.is_open())
    {
        getline(readFromFile, data);    //used to get whole line from file in the data
        cout << "Line " << ++i << " : " << data;

        while (!readFromFile.eof())    //reading while end of file is not
        {
            getline(readFromFile, data);    //used to get whole line from file in
            cout << "\nLine " << ++i << " : " << data;
        }
    }
    else
        cout << "file not open\n";

    readFromFile.close();
    return 0;
}
```



The screenshot shows a code editor with several tabs: Source1.cpp, file1.txt, L14_Q1.cpp, L13_Q4.cpp, L4_Q3.cpp, and L14_Q2.cpp. The 'file1.txt' tab is active, displaying the following text:

```
1 Off stream variable is used to write data in file
2 5
3 After using append
4
```

Below the code editor, a 'Microsoft Visual Studio Debug' window is open, showing the output of the program:

```
Line 1 : Off stream variable is used to write data in file
Line 2 : 5
Line 3 : After using append
Line 4 :
C:\Users\muham\OneDrive\Desktop\DS TASKS\x64\Debug\DS TASKS.
Press any key to close this window . . .|
```

Reading whole file method 2

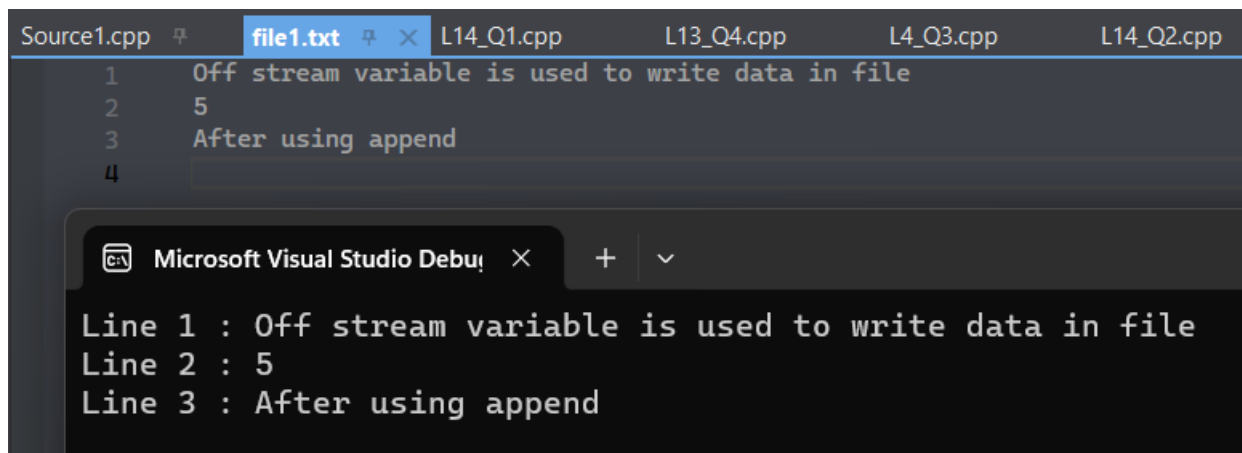
There is a little difference between both methods of reading. In this the last empty line is not read.

```
#include<iostream>
#include<string> // used to getline
#include<fstream> //include this to write in file
using namespace std;
int main()
{
    ifstream readFromFile;

    readFromFile.open("file1.txt"); //opening the file
    string data; //variable to store data
    int i = 0;
    if (readFromFile.is_open())
    {
        while (getline(readFromFile, data)) //reading while file is not end
        {
            cout << "Line " << ++i << " : " << data << endl;
        }
    }

    else
        cout << "file not open\n";

    readFromFile.close();
    return 0;
}
```



```
Source1.cpp  file1.txt  L14_Q1.cpp  L13_Q4.cpp  L4_Q3.cpp  L14_Q2.cpp
1  Off stream variable is used to write data in file
2  5
3  After using append
4
Microsoft Visual Studio Debug Console
Line 1 : Off stream variable is used to write data in file
Line 2 : 5
Line 3 : After using append
```

Getting data from file 1 and storing in file 2 after editing

```
#include<iostream>
#include<string> // used to getline
#include<fstream> //include this to write in file
using namespace std;
int main()
{
    ifstream readFromFile;
    ofstream writeInFile;

    readFromFile.open("file1.txt"); //opening the file to get data
    writeInFile.open("NewFile.txt", ios::app); //new file to write edited data

    string name; //variable to store data
    int marks;

    if (readFromFile.is_open() && writeInFile.is_open()) //checking if both files
are open
    {
        writeInFile << "Name Marks\n";
        while (readFromFile >> name >> marks) //reading from file1.txt while
file is not end
        {
            writeInFile << name << " " << marks + 5 << endl; //writing in new
file after by adding 5 in marks
        }
    }

    else
        cout << "file not open\n";

    //closing both file
    readFromFile.close();
    writeInFile.close();
    return 0;
}
```

Source1.cpp	file1.txt	Taha_22F-3277.cpp	NewFile.txt
1	Asad 48		1 Name Marks
2	Qasim 36		2 Asad 53
3	Ali 52		3 Qasim 41
			4 Ali 57
			5